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Course : ADVANCED WIRELESS COMMUNICATION SYSTEM

Exp No : 01

Simulation and Comparative Analysis of FDMA, TDMA and CDMA: Resource Allocation, User Capacity and Mobile-Generation Context

Aim:

To simulate and compare the basic resource-allocation methods used in FDMA, TDMA and CDMA, and examine how their assumed user capacity and interference behaviour differ.

Program code:(OBJECTIVE 1&2):

```
>> % Comparison of FDMA, TDMA and CDMA
% Simplified illustrative model; not a full cellular-network simulation.
clear; clc; close all;

% Illustrative system parameters
totalBandwidthHz = 1e6; % Total bandwidth: 1 MHz
fdmaChannelHz = 200e3; % FDMA channel width: 200 kHz
tdmaSlots = 8; % Number of TDMA slots per frame
processingGain = 64; % CDMA processing gain
targetSIRdB = 7; % Minimum acceptable CDMA SIR

% Capacity estimates under the stated assumptions
fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);
tdmaUsers = tdmaSlots;
targetSIR = 10^(targetSIRdB / 10);
cdmaUsers = floor(1 + processingGain / targetSIR);

fprintf('Illustrative maximum supported users\n');
fprintf('FDMA: %d users\n', fdmaUsers);
fprintf('TDMA: %d users\n', tdmaUsers);
fprintf('CDMA: %d users at a target SIR of %.1f dB\n\n', ...
    cdmaUsers, targetSIRdB);

% CDMA SIR as the number of simultaneous users increases
K = 2:30;
sirLinear = processingGain ./ (K - 1);
sirDb = 10 * log10(sirLinear);

% Display a comparison table
method = ["FDMA"; "TDMA"; "CDMA"];
supportedUsers = [fdmaUsers; tdmaUsers; cdmaUsers];
results = table(method, supportedUsers);
disp(results);

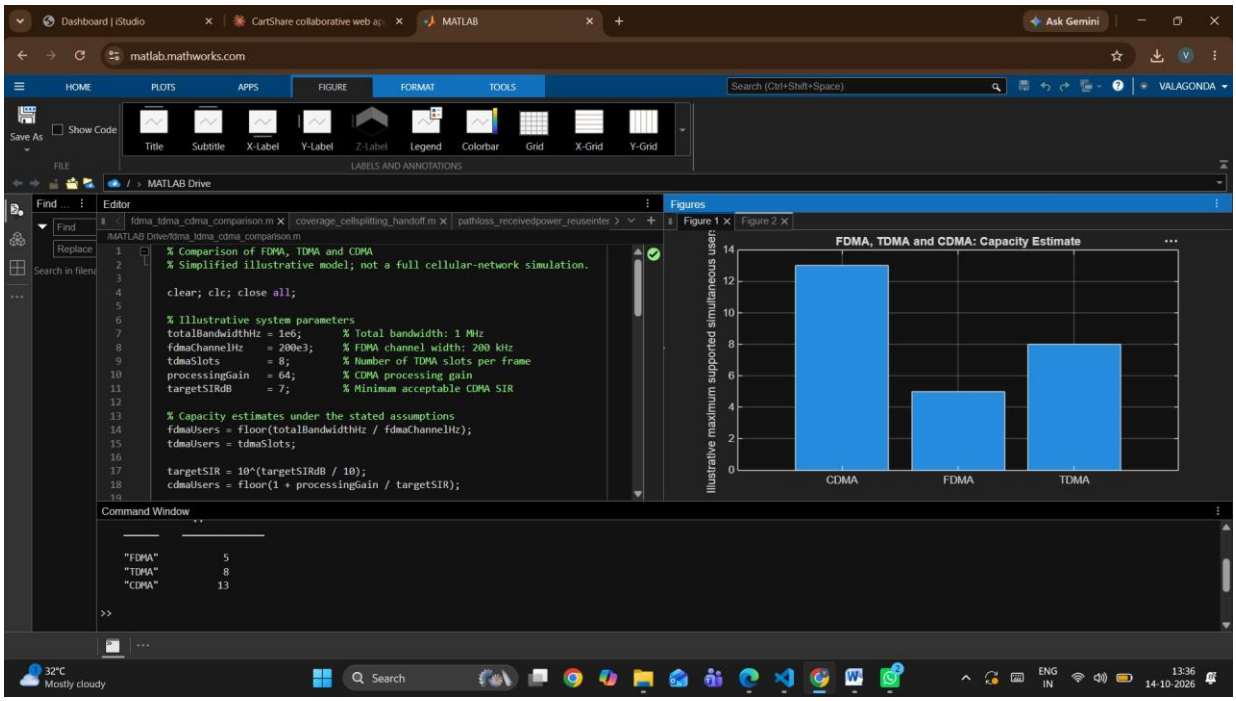
% Plot estimated user capacity
figure;
```

```
bar(categorical(method), supportedUsers);
ylabel('Illustrative maximum supported simultaneous users');
title('FDMA, TDMA and CDMA: Capacity Estimate');
grid on;

% Plot estimated CDMA SIR as users are added
figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5);
hold on;
yline(targetSIRdB, '--r', 'Target SIR');
xlabel('Number of simultaneous CDMA users (K)');
ylabel('Estimated SIR (dB)');
title('Estimated CDMA SIR versus Number of Users');
grid on;
hold off;
```

OBJECTIVE 1:

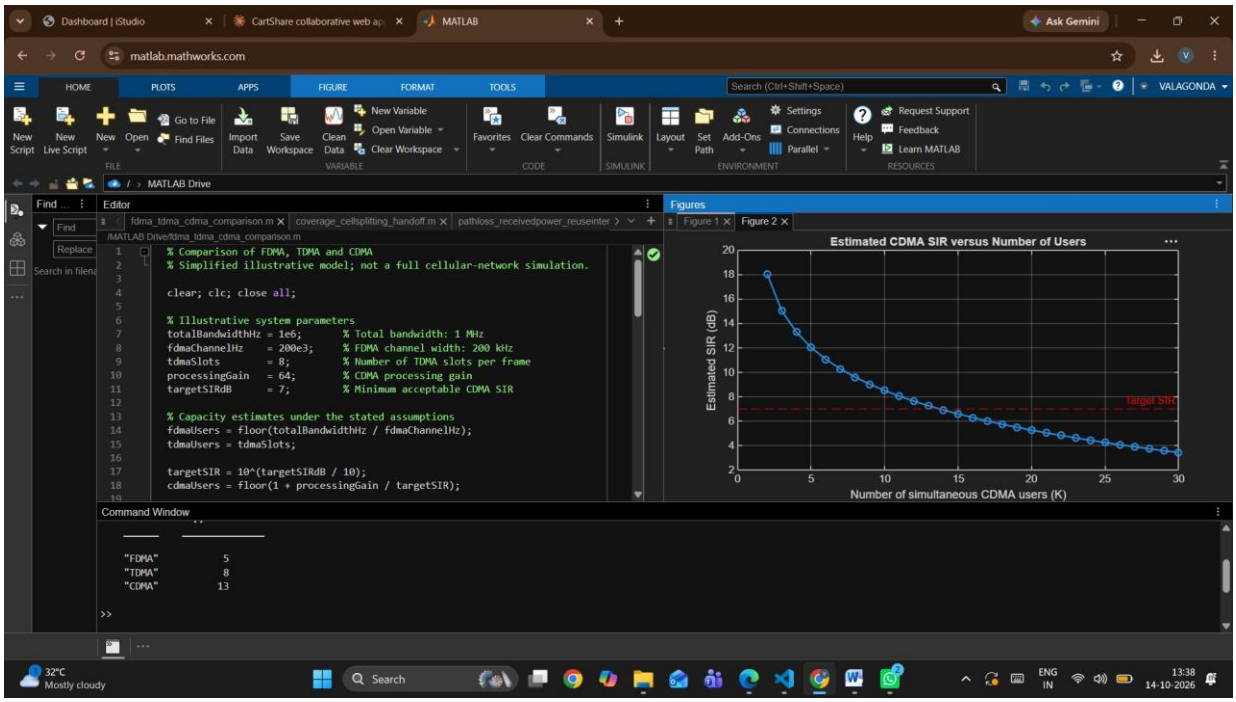
To estimate and compare the illustrative number of users supported by FDMA, TDMA and CDMA using the specified system parameters.



Output: FDMA = 5, TDMA = 8, CDMA = 13 users, and the capacity-estimate bar chart.

OBJECTIVE 2:

To analyze how the estimated CDMA signal-to-interference ratio changes as the number of simultaneous users increases.



Output: estimated CDMA SIR versus number of users, with the 7 dB target SIR line.

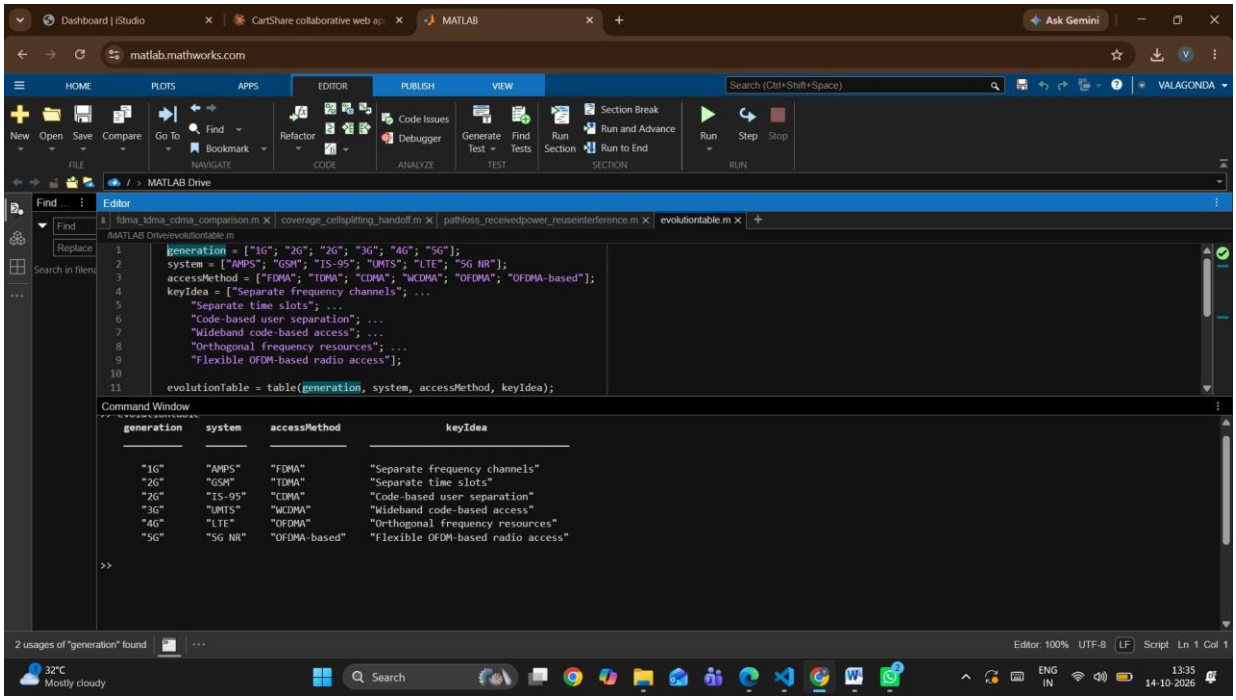
OBJECTIVE 3: Mobile-generation comparison

To relate representative multiple-access techniques to the evolution of mobile communication systems from 1G to 5G.

Program code:

```
generation = ["1G"; "2G"; "2G"; "3G"; "4G"; "5G"];
system = ["AMPS"; "GSM"; "IS-95"; "UMTS"; "LTE"; "5G NR"];
accessMethod = ["FDMA"; "TDMA"; "CDMA"; "WCDMA"; "OFDMA"; "OFDMA-based"];
keyIdea = ["Separate frequency channels"; ...
    "Separate time slots"; ...
    "Code-based user separation"; ...
    "Wideband code-based access"; ...
    "Orthogonal frequency resources"; ...
    "Flexible OFDM-based radio access"];

evolutionTable = table(generation, system, accessMethod, keyIdea);
disp(evolutionTable);
```



Output: evolution table relating generation, system, access method and key idea.

OBJECTIVE 4: Resource-allocation diagrams

To visualize how FDMA, TDMA and CDMA assign frequency, time and code resources to multiple users.

Code:

```
figure;
tiledlayout(1,3);

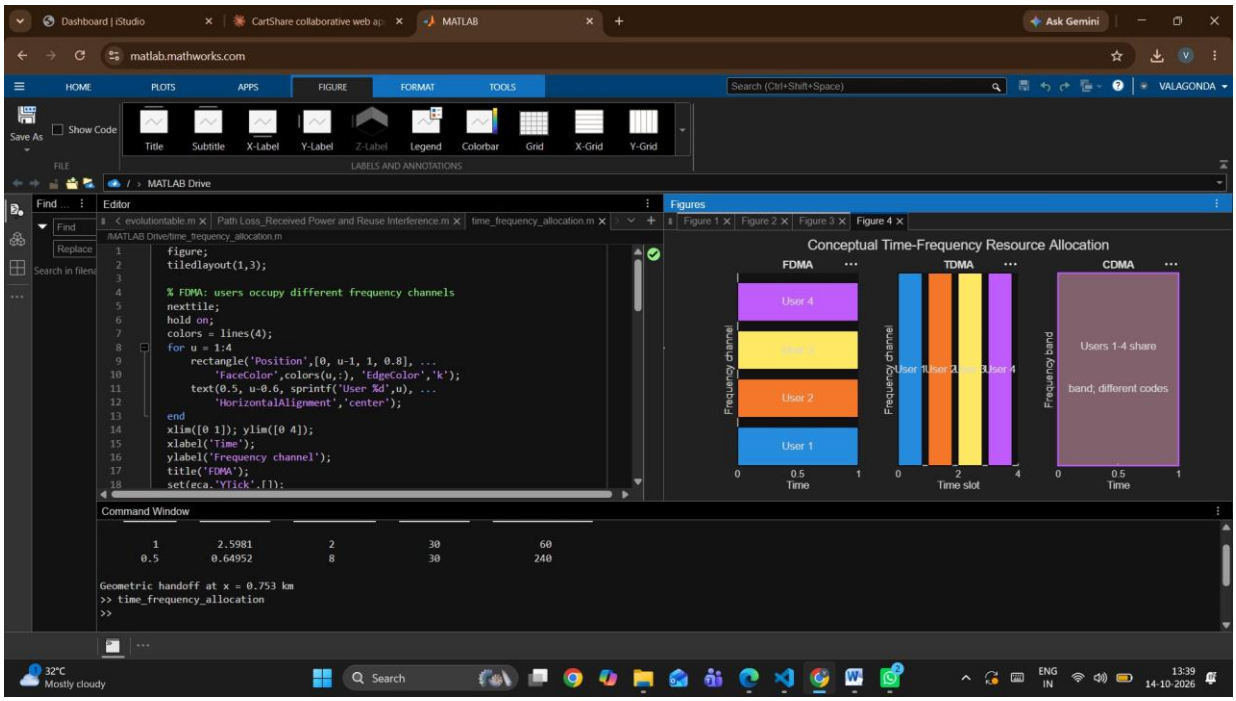
% FDMA: users occupy different frequency channels
nexttile;
hold on;
colors = lines(4);
for u = 1:4
    rectangle('Position',[0, u-1, 1, 0.8], ...
        'FaceColor',colors(u,:), 'EdgeColor','k');
    text(0.5, u-0.6, sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 1]); ylim([0 4]);
xlabel('Time');
ylabel('Frequency channel');
title('FDMA');
set(gca,'YTick',[]);
hold off;

% TDMA: users occupy different time slots on one frequency channel
nexttile;
hold on;
for u = 1:4
    rectangle('Position',[u-1, 0, 0.8, 1], ...
        'FaceColor',colors(u,:), 'EdgeColor','k');
```

```
    text(u-0.6, 0.5, sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 4]); ylim([0 1]);
xlabel('Time slot');
ylabel('Frequency channel');
title('TDMA');
set(gca,'YTick',[]);
hold off;

% CDMA: users share the same time and frequency band using different codes
nexttile;
hold on;
for u = 1:4
    rectangle('Position',[0, 0, 1, 1], ...
        'FaceColor',colors(u,:), 'FaceAlpha',0.22, ...
        'EdgeColor',colors(u,:), 'LineWidth',1.5);
end
text(0.5, 0.62, 'Users 1-4 share', 'HorizontalAlignment','center');
text(0.5, 0.40, 'band; different codes', 'HorizontalAlignment','center');
xlim([0 1]); ylim([0 1]);
xlabel('Time');
ylabel('Frequency band');
title('CDMA');
set(gca,'YTick',[]);
hold off;

sgtitle('Conceptual Time-Frequency Resource Allocation');
```



Output: conceptual time-frequency resource allocation for FDMA, TDMA and CDMA.

Result:

FDMA, TDMA and CDMA were simulated in MATLAB with the stated illustrative parameters. FDMA supported 5 users (1 MHz / 200 kHz), TDMA supported 8 users (one per slot) and CDMA supported 13 users at a 7 dB target SIR. The CDMA SIR fell as simultaneous users increased, confirming that CDMA capacity is interference-limited. The evolution table linked FDMA, TDMA, CDMA, WCDMA and OFDMA-based access to 1G through 5G.